# Morris Alper, Ph.D.

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I research creative multimodal machine learning inspired by linguistics—how structures encoded by language provide a tool for interpreting and controlling modalities such as vision and audio, ML as a creativity aid, and innovative methods for low-resource languages.

## Education

**Ph.D., Tel Aviv University** 2023 – 2025
Electrical Engineering (Machine Learning)
Advisors: *Dr. Hadar Averbuch-Elor*, *Dr. Raja Giryes*
Dissertation: *Multimodal Learning for High-Level Semantic Understanding: From Interpretability to Real-World Applications*

**M.Sc. (cum laude), Tel Aviv University** 2019 – 2023
Computer Science
Advisor: *Dr. Hadar Averbuch-Elor*
Thesis: *Learning Human-Human Interactions in Images with Weak Textual Supervision*
GPA: 98 / 100

**B.Sc., Massachusetts Institute of Technology (MIT)** 2012 – 2016
Linguistics and Mathematics (dual major). GPA: 4.9 / 5

## Employment History

**Assistant Professor,** Department of Computer Science, University of Miami 2026 – [ongoing]

**Research Analyst (Postdoctoral),** Carnegie Mellon University 2025 – 2026
Advisor: *Dr. Lori Levin*
Developing machine learning methods for endangered language documentation.

**Research Scientist Intern,** Meta 2024
Developing 3D generative models using weakly-structured image collections.

**Data Science Lead and Lecturer,** Israel Tech Challenge 2019 – 2023
Designed syllabus, wrote, and delivered lectures on data science to industry professionals.

**NLP Data Scientist,** Pipl, Inc. 2019 – 2021
Designed information extraction algorithms for unstructured textual web data.

**NLP Data Scientist,** Reverso-Softissimo 2018 – 2019
Designed NLP algorithms to automatically correct and enrich linguistic data.

**Software Developer,** K Dictionaries 2017 – 2018
Managed API for lexicographic data, and developed algorithms for data enrichment.

**Algorithm Development Intern,** SECOM 2016
Developed algorithms for improved facial recognition.

**Software and Algorithm Developer,** Voiceitt 2015 – 2016
Prototyped speech recognition algorithms and integrated them into the company's API.

**Undergraduate Research in Cognitive Linguistics,** TedLab, MIT 2012–2015
Analyzed psycholinguistic survey data to study cognition and language.

## Research Publications

* indicates equal contribution.

### Peer-Reviewed Conference Papers

1. Maxim Melichov*, Yakov Kolani*, **Morris Alper**. ReNikud: Audio-Supervised Hebrew Grapheme-to-Phoneme Conversion. IEEE SLT, 2026.
2. **Morris Alper***, Enora Rice*, Bhargav Shandilya*, Alexis Palmer, Lori Levin. CWoMP: Morpheme Representation Learning for Interlinear Glossing. EMNLP 2026. *(Oral)*
3. Yakov Kolani, Maxim Melichov, Cobi Calev, **Morris Alper**. Phonikud: Overcoming Phonetic Underspecification for Hebrew Text-To-Speech. Interspeech, 2026.
4. **Morris Alper***, Moran Yanuka*, Raja Giryes, Gašper Beguš. ConlangCrafter: Constructing Languages with a Multi-Hop LLM Pipeline. *ACL*, 2026. *(Oral)*
5. **Morris Alper**, David Novotny, Filippos Kokkinos, Hadar Averbuch-Elor, Tom Monnier. WildCAT3D: Appearance-Aware Multi-View Diffusion in the Wild. *NeurIPS*, 2025.
6. Rachel Mikulinsky*, **Morris Alper***, Shai Gordin, Enrique Jiménez, Yoram Cohen, Hadar Averbuch-Elor. ProtoSnap: Prototype Alignment For Cuneiform Signs. *ICLR*, 2025.
7. Keren Ganon*, **Morris Alper***, Rachel Mikulinsky, Hadar Averbuch-Elor. WAFFLE: Multimodal Floorplan Understanding in the Wild. *WACV*, 2025. *(Oral)*
8. **Morris Alper** and Hadar Averbuch-Elor. Emergent Visual-Semantic Hierarchies in Image-Text Representations. *ECCV*, 2024. *(Oral)*
9. Assaf Ben-Kish, Moran Yanuka, **Morris Alper**, Raja Giryes, Hadar Averbuch-Elor. Mitigating Open-Vocabulary Caption Hallucinations. *EMNLP*, 2024.
10. Moran Yanuka, **Morris Alper**, Hadar Averbuch-Elor, Raja Giryes. ICC: Quantifying Image Caption Concreteness for Multimodal Dataset Curation. *ACL (Findings)*, 2024.
11. Chen Dudai*, **Morris Alper***, Hana Bezalel, Rana Hanocka, Itai Lang, and Hadar Averbuch-Elor. HaLo-NeRF: Learning Geometry-Guided Semantics for Exploring Unconstrained Photo Collections. *Eurographics*, 2024.
12. **Morris Alper** and Hadar Averbuch-Elor. Kiki or Bouba? Sound Symbolism in Vision-and-Language Models. *NeurIPS*, 2023. *(Spotlight)*
13. **Morris Alper** and Hadar Averbuch-Elor. Learning Human-Human Interactions in Images from Weak Textual Supervision. *ICCV*, 2023.
14. **Morris Alper***, Michael Fiman*, and Hadar Averbuch-Elor. Is BERT Blind? Exploring the Effect of Vision-and-Language Pretraining on Visual Language Understanding. *CVPR*, 2023.

### Peer-Reviewed Workshop Papers

1. Ben Sapirstein, Roy Mattar, Guy Mor-Lan, Ahlam Mohamed, Letizia Cerqueglini, **Morris Alper**. SHAMS: An Audio-Grounded Pronunciation Benchmark for Levantine Arabic. *ArabicNLP*, 2026.
2. Shai Gordin, **Morris Alper**, Avital Romach, Luis Saenz Santos, Naama Yochai, Roey Lalazar. CuReD: Deep Learning OCR for Cuneiform Text Editions and Legacy Materials. *ML4AL Workshop*, 2024. (Oral)

### Additional Workshop Papers

1. **Morris Alper**, Samopriya Basu, Nathan Bennett, Ryan Kessler, S Verlaine Ravana, Wendy F. K'ah Skáahluwaa Todd. Towards Low-Resource Text-to-Speech Generation for Indigenous Pacific Northwest Languages: The Case of Haida Xaad Kíl. *ICSNL*, 2025.
2. **Morris Alper**. Auto-generating Bilingual Dictionaries: Results of the TIAD-2017 Shared Task Baseline Algorithm. *LDK Workshops*, 2017.

### Preprints and Under Review

1. Moran Yanuka, Raja Giryes, **Morris Alper**. Phonological Interference in Multilingual Speech Models. 2026.
2. Candace Makeda Moore, **Morris Alper**. Benchmarking Open-Source Automated Thigh Muscle MRI Segmentation Algorithms. 2026.
3. **Morris Alper**, Vasudha Varadarajan, Moran Yanuka, Angelina Wang, Hadar Averbuch-Elor. NAME-SAKES: Probing Identity Memorization in Text-to-Image Models. 2026.
4. Shahaf Pruss*, **Morris Alper***, Hadar Averbuch-Elor. Dynamic Scene Understanding from Vision-Language Representations. 2025.

## Teaching Experience

**Introduction to Artificial Intelligence (CSC345)**, Instructor, University of Miami Fall 2026 [ongoing]

**Computer Vision for Material Culture (Digital Humanities)**, DANES ScHack 2026 Summer School, Turin, Italy July 2026

**Human Language for Artificial Intelligence**, Teaching Assistant, Carnegie Mellon University Fall 2025

**Digital Signal Processing**, Teaching Assistant, Tel Aviv University Fall 2023

**Data Science Lead and Lecturer**, Israel Tech Challenge 2019 – 2023

- Designed syllabus spanning fundamentals through advanced topics (deep learning, NLP, computer vision)
- Lectured on data science and machine learning to diverse industry professionals across multiple cohorts
- Mentored student capstone projects in technical implementation and communicating results
- Achieved strong student evaluations (consistently above 6.0 on 7-point scale)

## Grants and Funding

*Awarded*

**Paleography Powered by Machine Learning: Analyzing Cuneiform Documents with Neural Networks** 2023
LMU-TAU Research Cooperation Program
Role: Grant co-writer and funded researcher (PIs: Yoram Cohen, Hadar Averbuch-Elor, Enrique Jiménez)

*Pending*

**Deep Research Agents for Archaeology and Cultural Heritage** 2026
Humanities and Artificial Intelligence Virtual Institute (HAVI; Schmidt Sciences)
Role: Co-PI (Lead PI: Daphne Ippolito)

**Tracing the Signs: Multimodal Foundation Models for Ancient Writing** 2026
Level I Grant, Humanities and Artificial Intelligence Virtual Institute (HAVI; Schmidt Sciences)
Role: Co-PI (Lead PI: Ahmad Al-Jallad)

**Collaborative Research: RI: Beyond IPA: Scalable Multilingual Speech Technology Without Symbolic Supervision** 2026
National Science Foundation (NSF), Robust Intelligence (RI)
Role: Lead PI (co-PI: Shinji Watanabe, CMU)

**Collaborative Research: NSF-BSF: RI: Reading and Writing Between the Lines: Bridging Rasterized and Parametric Foundation Models for Visual Text** 2026
National Science Foundation (NSF) and Binational Science Foundation (BSF), Robust Intelligence (RI)
Role: Lead PI (co-PI: Taylor Berg-Kirkpatrick, UCSD; Israeli PI: Shai Gordin, Ariel University)

## Talks and Presentations

**Invited Talks (Selected)**

2026 **Sound, Script, and Syntax in Multimodal AI**. Technion - Israel Institute of Technology.
2026 **Effective Collaboration as Bilingualism: Lessons from AI-Humanities Research**. Archaeology, Ancient History, and AI Workshop, Johns Hopkins University.
2026 **ConlangCrafter: Constructing Languages with a Multi-Hop LLM Pipeline**. Workshop on Invented, Constructed, and Emergent Languages in Multi-AI Systems, MIT.
2026 **CWoMP: Morpheme Representation Learning for Interlinear Glossing**. NLP Seminar, National Research Council Canada.
2026 **Phonikud: Overcoming Phonetic Underspecification for Hebrew Text-To-Speech**. WAVLab Seminar, CMU.
2026 **Old Words, New Words: ML for Ancient and Fictional Languages**. Guest Lecture for Linguistics 265: Biological and Artificial Language, UC Berkeley.
2025 **Old Words, New Words: ML for Ancient and Fictional Languages**. IBM Israel.
2025 **Understanding the Knowledge of Vision-Language Models**. SC Lab Seminar, UC Berkeley.
2025 **Emergent Visual-Semantic Hierarchies in Image-Text Representations**. NYC Computer Vision Day 2025.
2024 **Understanding the Knowledge of Vision-Language Models**. IMAGINE Lab Seminar, Ecole Nationale des Ponts et Chaussées.
2024 **Emergent Visual-Semantic Hierarchies in Image-Text Representations**. Vision & AI Seminar, Weizmann Institute of Science.
2024 **Kiki or Bouba? Sound Symbolism in Vision-and-Language Models**. Israel Machine Vision Conference (IMVC).
2023 **What is Learned by Vision and Language Models**. School of Electrical and Computer Engineering, Ben Gurion University.
2023 **What is Learned by Vision and Language Models**. Shevah Mofet High School Linguistics Club.
2023 **Is BERT Blind? Exploring the Effect of Vision-and-Language Pretraining on Visual Language Understanding**. Vision Seminar, Hebrew University of Jerusalem.
2023 **"Deeplomatics": Deep learning for Automatic Analysis of Cuneiform Texts**. First Meeting of the Digital Ancient Near Eastern Studies Network (DANES).
2022 **"Deeplomatics": Deep learning for Automatic Analysis of Cuneiform Texts**. LMU-TAU Workshop for Early-Career Researchers.
2017 **Auto generating bilingual dictionaries**. eLex 2017 conference, Leiden, the Netherlands.

**Industry Workshops** (delivered through Israel Tech Challenge)

2021–2022 **NLP Workshop Series**. AT&T Israel.
2020 **Cutting-Edge NLP With Transformers and BERT.** PayPal Israel.

**Press Coverage**

2026 ConlangCrafter featured in Science, IEEE Spectrum, University of Miami News, Sun Sentinel, The Miami Hurricane, and NBC6 South Florida [video]
2025 ProtoSnap featured in Cornell News
2024 Kiki-Bouba sound symbolism work featured in Haaretz (Hebrew)
2024 Radio interview for Kan Tarbut program on VLM sound symbolism research (46:30-60:00, Hebrew language) [audio]

**Recorded Presentations**

2024 IMVC 2024 presentation on "Kiki or Bouba?" NeurIPS 2023 paper [video]
2023 "Deeplomatics" presentation at DANES [video]
2017 "Auto generating bilingual dictionaries" at eLex [video]

## Service

### Peer Review Service

2026 ACL ARR (Jan, Mar, May), CoLM, Humanities and Social Sciences Communications [ongoing]
2025 SIGGRAPH, NeurIPS *(Top Reviewer)*, ICLR, ACL ARR
2024 COLM, NeurIPS, ACML, ICLR

### Committees

2027 ALP (Ancient Language Processing) Workshop, Co-Organizer; EvaCun Shared Task, Co-Lead *(pending)*

## Awards and Achievements

2011 **International Linguistics Olympiad**, first place internationally (both individual and team rounds), best solution (problem 2), Zhurinsky Memorial Prize.
2009 **International Linguistics Olympiad**, honorable mention (individual round), first place in team round.
2008 **International Linguistics Olympiad**, tied for 6th place individual round (silver medal).

## Languages

**English** (native), **Hebrew** (professional), **French** (upper intermediate)